INTERNET PROGRAMMING LAB

Experiment No. 8

Title: Hospital Online Appointment Booking System (MedCare Hospital)

AIM

To design and develop an online hospital appointment booking web application using HTML, CSS and JavaScript that allows patients to sign up, search doctors by specialty and location, book and cancel appointments, view appointment history and rate doctors, and allows the administrator to manage doctors and view appointment statistics.

ALGORITHM

Step 1. Start the application and display the Patient Signup / Login page.

Step 2. Signup: read username, password and 10-digit mobile number; validate and store the patient.

Step 3. Login: verify credentials; patients go to the Search page and the administrator goes to the Admin Dashboard.

Step 4. Search Module: read the selected specialty and location, filter the doctor list and display name, specialty, hospital, average rating and a Google Maps link.

Step 5. On clicking Book, ensure the user is logged in and open the booking page with a date picker and time slots.

Step 6. Show time slots for the selected date; slots already booked with the same doctor are disabled.

Step 7. Read the selected slot and reason for visit, create the appointment with status "Booked" and show a confirmation (reminder by e-mail/SMS).

Step 8. My Appointments: list the patient's appointments; allow Cancel (status "Cancelled") or Mark visited (status "Completed").

Step 9. For completed appointments allow the patient to rate the doctor (1-5); recompute the doctor's average rating.

Step 10. Admin Dashboard: compute total, booked, completed and cancelled appointments and number of doctors; list doctors with ratings and appointment counts.

Step 11. Admin may add a new doctor or delete an existing doctor, and view all appointments.

Step 12. Logout and stop.

PROGRAM

File: p8.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>MedCare Hospital - Online Appointment Booking</title>
<style>
:root{--c:#0f7b6c}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.slot{display:inline-block;border:1px solid #0f7b6c;border-radius:5px;padding:6px
10px;margin:3px;cursor:pointer;font-size:13px}
.slot.sel{background:#0f7b6c;color:#fff}
.slot.off{background:#ddd;border-color:#bbb;color:#888;cursor:not-allowed}
.star{color:#f5a623}
</style>
</head>
<body>
<header>
  <h2>&#10010; MedCare Hospital</h2>
  <nav>
    <button onclick="go('search')">Search</button>
    <button onclick="go('mine')">My Appointments</button>
    <button onclick="go('admin')">Admin</button>
    <button id="authBtn" onclick="authClick()">Login</button>
  </nav>
</header>
<main>
  <!-- SIGNUP / LOGIN -->
  <section id="login" class="page card">
    <h3>Patient Signup / Login</h3>
    <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
    <input id="lph" placeholder="Mobile (for SMS reminder)">
    <button class="btn" onclick="login()">Login</button>
    <button class="btn gray" onclick="signup()">Signup</button>
```

```
    <p id="lmsg"></p><small>Admin demo login: admin / admin123</small>
  </section>

  <!-- SEARCH MODULE -->
  <section id="search" class="page card">
    <h3>Search Doctor / Hospital</h3>
    Specialty <select id="sp"></select>
    Location <select id="loc"></select>
    <button class="btn" onclick="doSearch()">Search</button>
    <div id="res" class="grid" style="margin-top:12px"></div>
  </section>

  <!-- APPOINTMENT BOOKING -->
  <section id="book" class="page card">
    <h3>Book Appointment</h3>
    <div id="dinfo"></div>
    Date <input type="date" id="bd" onchange="drawSlots()">
    <div id="slots" style="margin:8px 0"></div>
    <input id="reason" placeholder="Reason / symptoms" style="width:60%">
    <button class="btn" onclick="confirmBook()">Confirm Appointment</button>
    <p id="bmsg"></p>
  </section>

  <!-- APPOINTMENT MANAGEMENT -->
  <section id="mine" class="page card">
    <h3>My Appointments (status &amp;amp; history)</h3>
    <div id="mlist"></div>
  </section>

  <!-- ADMIN -->
  <section id="admin" class="page card">
    <h3>Admin Dashboard</h3>
    <div id="stats"></div>
    <h3 style="margin-top:12px">Manage Doctors</h3>
    <input id="dn" placeholder="Doctor name"><input id="ds" placeholder="Specialty">
    <input id="dh" placeholder="Hospital"><input id="dl" placeholder="Location">
    <button class="btn" onclick="addDoc()">Add Doctor</button>
    <p id="amsg"></p>
    <div id="dtable"></div>
    <h3 style="margin-top:12px">Appointments</h3><div id="atable"></div>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

let users = load('mc_users', [{u:'admin', p:'admin123', role:'admin'}]);
let docs  = load('mc_docs', [
  {id:1, name:'Dr. Anita Rao',    spec:'Cardiology',  hosp:'MedCare Heart Centre', loc:'Chennai',   ratings:
[5,4,5]},
  {id:2, name:'Dr. Suresh Kumar', spec:'Orthopedics', hosp:'MedCare Central',      loc:'Chennai',   ratings:
[4,4]},
  {id:3, name:'Dr. Priya Nair',   spec:'Pediatrics',  hosp:'MedCare Children',     loc:'Coimbatore',ratings:
[5,5,4]},
  {id:4, name:'Dr. Imran Khan',   spec:'Dermatology', hosp:'MedCare Skin Clinic',  loc:'Madurai',   ratings:
[4]},
  {id:5, name:'Dr. Lakshmi Iyer', spec:'Cardiology',  hosp:'MedCare Central',      loc:'Coimbatore',ratings:
[5,4]}
]);
let appts = load('mc_appts', []);
let me = null, curDoc = null, curSlot = null;
const SLOTS = ['09:00','09:30','10:00','10:30','11:00','11:30','14:00','14:30','15:00','15:30'];

function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }
function go(p) {
  if (['mine', 'book'].includes(p) && !me) { msg('lmsg', 'Please login first.', 1); p = 'login'; }
```

```
  if (p === 'admin' && (!me || me.role !== 'admin')) { msg('lmsg', 'Admin login required.', 1); p = 'login';
}
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on'));
  $(p).classList.add('on');
  if (p === 'mine') renderMine();
  if (p === 'admin') renderAdmin();
}
const avg = d => d.ratings.length ? (d.ratings.reduce((a, b) => a + b, 0) / d.ratings.length).toFixed(1) :
'New';

/* ---------- signup / login ---------- */
function signup() {
  const u = $('lu').value.trim(), p = $('lp').value, ph = $('lph').value.trim();
  if (!u || p.length < 4 || !/^\d{10}$/.test(ph)) return msg('lmsg', 'Enter username, password (4+ chars)
and 10-digit mobile.', 1);
  if (users.some(x => x.u === u)) return msg('lmsg', 'Username already taken.', 1);
  users.push({u, p, ph, role:'patient'}); save('mc_users', users); msg('lmsg', 'Signup successful. Please
login.');
}
function login() {
  const f = users.find(x => x.u === $('lu').value.trim() && x.p === $('lp').value);
  if (!f) return msg('lmsg', 'Invalid credentials.', 1);
  me = f; $('authBtn').textContent = 'Logout (' + f.u + ')'; msg('lmsg', 'Welcome ' + f.u);
  go(f.role === 'admin' ? 'admin' : 'search');
}
function authClick() { if (me) { me = null; $('authBtn').textContent = 'Login'; } go('login'); }

/* ---------- search ---------- */
function initSearch() {
  const specs = ['All', ...new Set(docs.map(d => d.spec))], locs = ['All', ...new Set(docs.map(d =>
d.loc))];
  $('sp').innerHTML = specs.map(s => `<option>${s}</option>`).join('');
  $('loc').innerHTML = locs.map(s => `<option>${s}</option>`).join('');
}
function doSearch() {
  const s = $('sp').value, l = $('loc').value;
  const list = docs.filter(d => (s === 'All' || d.spec === s) && (l === 'All' || d.loc === l));
  $('res').innerHTML = list.length ? list.map(d => `<div class="item"><div
class="pic">&#128104;&#8205;&#9877;&#65039;</div>
    <b>${d.name}</b><br><span class="pill">${d.spec}</span><br>${d.hosp}, ${d.loc}<br>
    <span class="star">&#9733;</span> ${avg(d)} (${d.ratings.length} reviews)<br>
    <button class="btn" onclick="openBook(${d.id})">Book</button>
    <a href="https://www.google.com/maps/search/?api=1&query=${encodeURIComponent(d.hosp + ' ' + d.loc)}"
target="_blank">View map</a></div>`).join('')
    : '<p class="err">No doctors found for the selected criteria.</p>';
}

/* ---------- booking ---------- */
function openBook(id) {
  if (!me) { msg('lmsg', 'Please login to book an appointment.', 1); return go('login'); }
  curDoc = docs.find(d => d.id === id); curSlot = null;
  $('dinfo').innerHTML = `<p><b>${curDoc.name}</b> - ${curDoc.spec}, ${curDoc.hosp} (${curDoc.loc})</p>`;
  const t = new Date(); t.setDate(t.getDate() + 1); $('bd').min = new Date().toISOString().slice(0, 10);
  $('bd').value = t.toISOString().slice(0, 10); $('bmsg').textContent = ''; drawSlots(); go('book');
}
function drawSlots() {
  const busy = appts.filter(a => a.did === curDoc.id && a.date === $('bd').value && a.status ===
'Booked').map(a => a.time);
  $('slots').innerHTML = SLOTS.map(s => busy.includes(s) ? `<span class="slot off">${s}</span>`
    : `<span class="slot ${s === curSlot ? 'sel' : ''}"
onclick="curSlot='${s}';drawSlots()">${s}</span>`).join('');
}
function confirmBook() {
  if (!curSlot) return msg('bmsg', 'Select a time slot.', 1);
  if (!$('reason').value.trim()) return msg('bmsg', 'Enter reason for visit.', 1);
  const a = {id:Date.now(), user:me.u, did:curDoc.id, doc:curDoc.name, hosp:curDoc.hosp, date:$('bd').value,
time:curSlot, reason:$('reason').value, status:'Booked', rated:false};
  appts.push(a); save('mc_appts', appts);
```

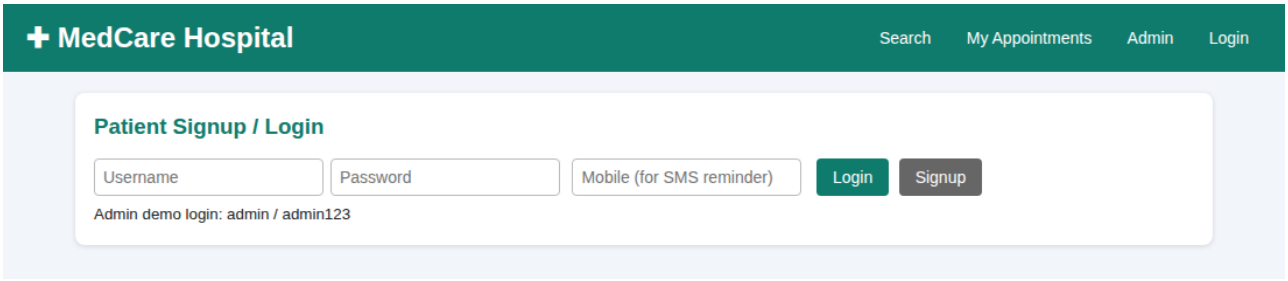
```
  msg('bmsg', `Appointment confirmed with ${a.doc} on ${a.date} at ${a.time}. Reminder will be sent by e-
mail/SMS.`);
  $('reason').value = ''; curSlot = null; drawSlots();
}

/* ---------- my appointments ---------- */
function renderMine() {
  const mine = appts.filter(a => a.user === me.u);
  $('mlist').innerHTML = mine.length ? '<table><tr><th>Doctor</th><th>Hospital</th><th>Date</th><th>Time</
th><th>Status</th><th>Action</th></tr>' +
    mine.map(a => `<tr><td>${a.doc}</td><td>${a.hosp}</td><td>${a.date}</td><td>${a.time}</td><td>$
{a.status}</td><td>
      ${a.status === 'Booked' ? `<button class="btn red" onclick="cancelA(${a.id})">Cancel</button>
      <button class="btn gray" onclick="complete(${a.id})">Mark visited</button>` : ''}
      ${a.status === 'Completed' && !a.rated ? `<select
id="r${a.id}"><option>5</option><option>4</option><option>3</option><option>2</option><option>1</option></
select>
      <button class="btn" onclick="rate(${a.id})">Rate</button>` : ''}</td></tr>`).join('') + '</table>' :
'<p>No appointments yet.</p>';
}
function cancelA(id) { appts.find(a => a.id === id).status = 'Cancelled'; save('mc_appts', appts);
renderMine(); }
function complete(id) { appts.find(a => a.id === id).status = 'Completed'; save('mc_appts', appts);
renderMine(); }
function rate(id) {
  const a = appts.find(x => x.id === id); docs.find(d => d.id === a.did).ratings.push(+$('r' + id).value);
  a.rated = true; save('mc_docs', docs); save('mc_appts', appts); renderMine();
}

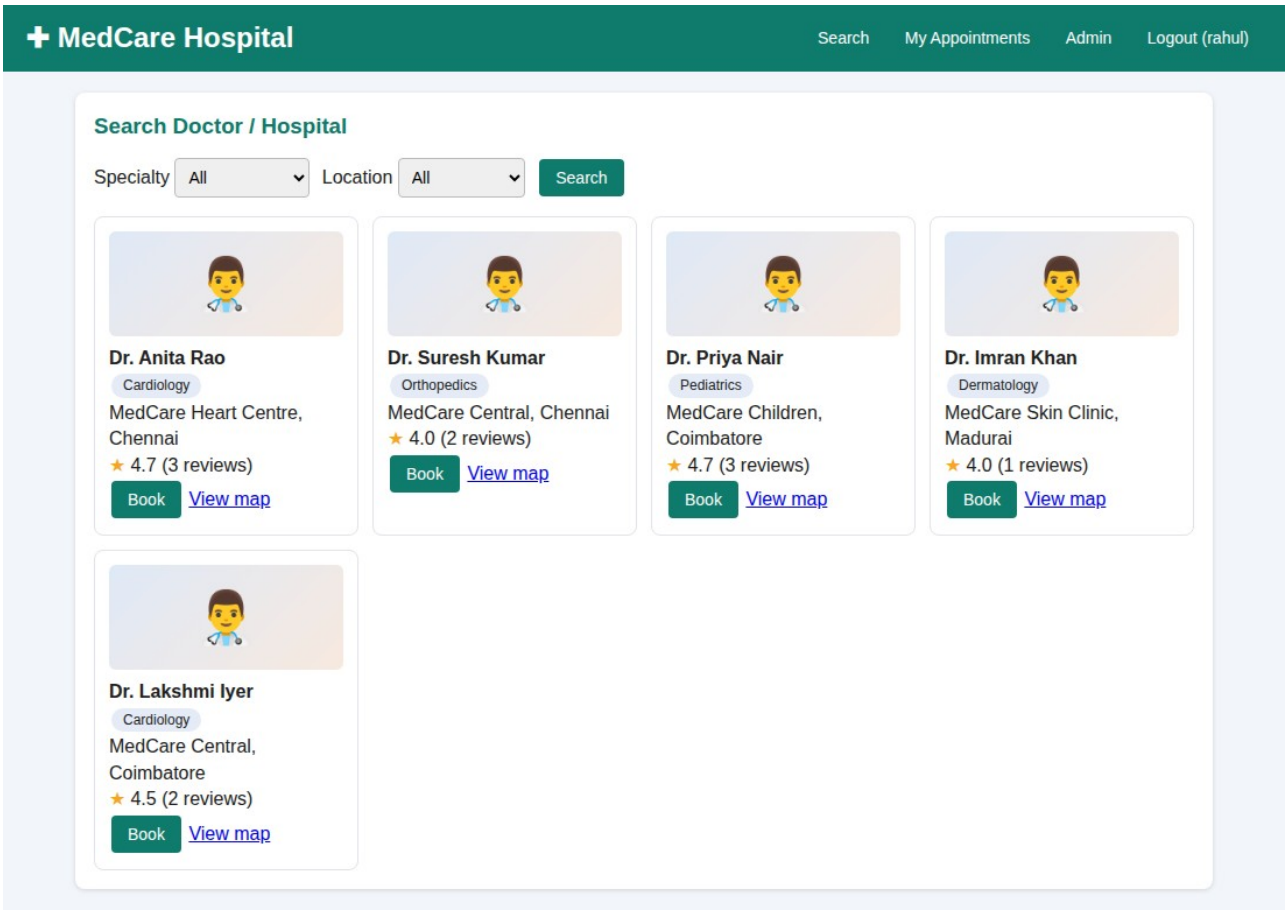
/* ---------- admin ---------- */
function renderAdmin() {
  const c = s => appts.filter(a => a.status === s).length;
  $('stats').innerHTML = `<div class="stat"><b>${appts.length}</b>Total</div><div class="stat"><b>$
{c('Booked')}</b>Booked</div>
    <div class="stat"><b>${c('Completed')}</b>Completed</div><div
class="stat"><b>${c('Cancelled')}</b>Cancelled</div><div class="stat"><b>${docs.length}</b>Doctors</div>`;
  $('dtable').innerHTML = '<table><tr><th>Doctor</th><th>Specialty</th><th>Hospital</th><th>Location</
th><th>Rating</th><th>Appts</th><th></th></tr>' +
    docs.map(d => `<tr><td>${d.name}</td><td>${d.spec}</td><td>${d.hosp}</td><td>${d.loc}</td><td>$
{avg(d)}</td>
    <td>${appts.filter(a => a.did === d.id).length}</td><td><button class="btn red" onclick="delDoc($
{d.id})">Delete</button></td></tr>`).join('') + '</table>';
  $('atable').innerHTML = appts.length ? '<table><tr><th>Patient</th><th>Doctor</th><th>Date</th><th>Time</
th><th>Status</th></tr>' +
    appts.map(a => `<tr><td>${a.user}</td><td>${a.doc}</td><td>${a.date}</td><td>${a.time}</td><td>$
{a.status}</td></tr>`).join('') + '</table>' : '<p>No appointments.</p>';
}
function addDoc() {
  const d = {id:Date.now(), name:$('dn').value.trim(), spec:$('ds').value.trim(), hosp:$('dh').value.trim(),
loc:$('dl').value.trim(), ratings:[]};
  if (!d.name || !d.spec || !d.hosp || !d.loc) return msg('amsg', 'Fill all doctor fields.', 1);
  docs.push(d); save('mc_docs', docs); initSearch(); renderAdmin(); msg('amsg', 'Doctor added.');
}
function delDoc(id) { docs = docs.filter(d => d.id !== id); save('mc_docs', docs); initSearch();
renderAdmin(); }

initSearch(); doSearch(); go('login');
</script>
</body>
</html>
```

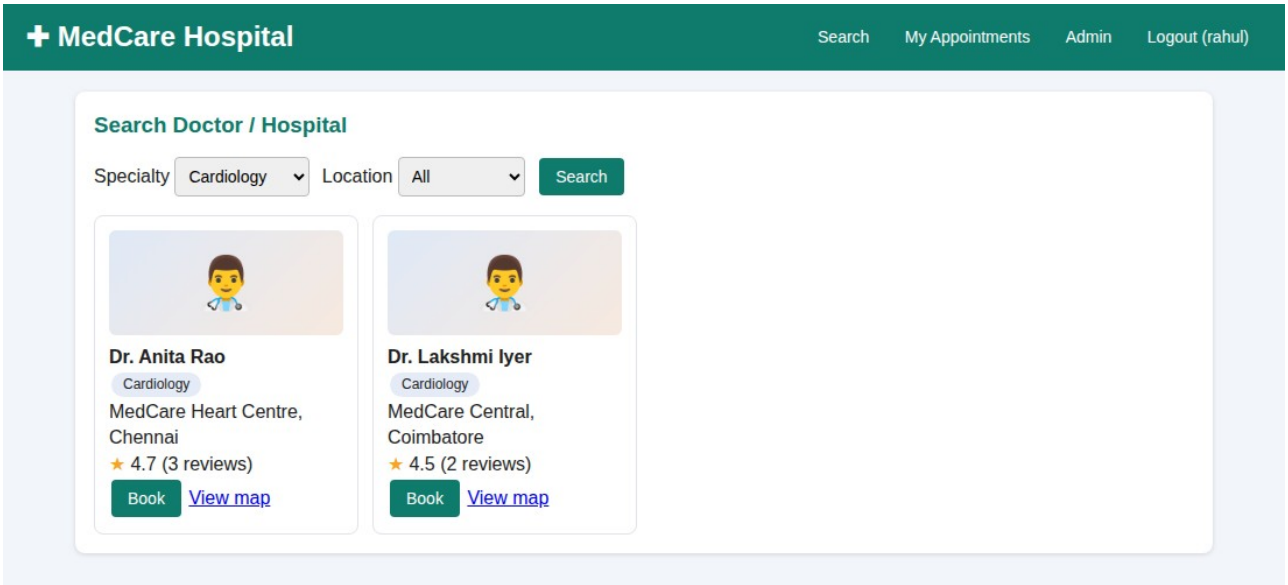
OUTPUT



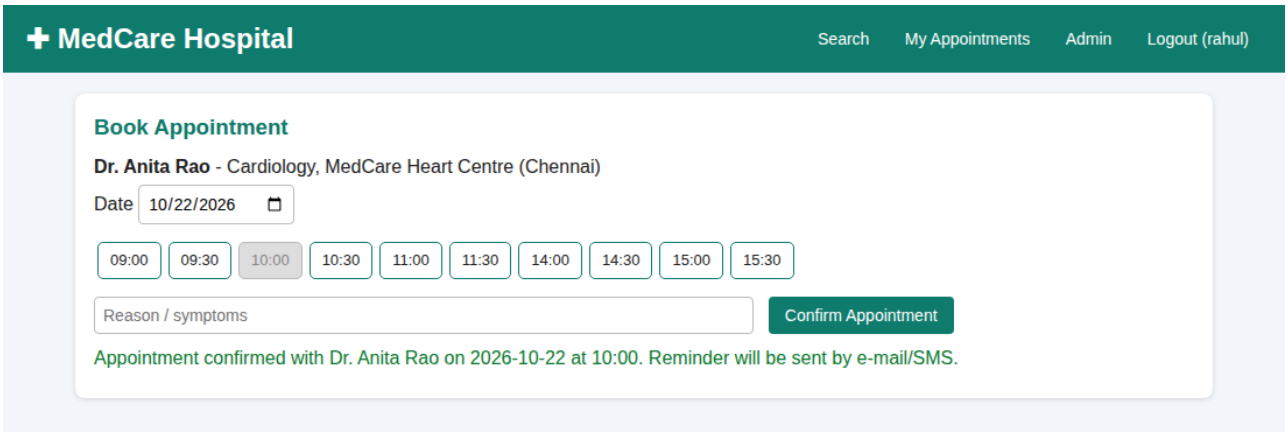
Patient Signup / Login page



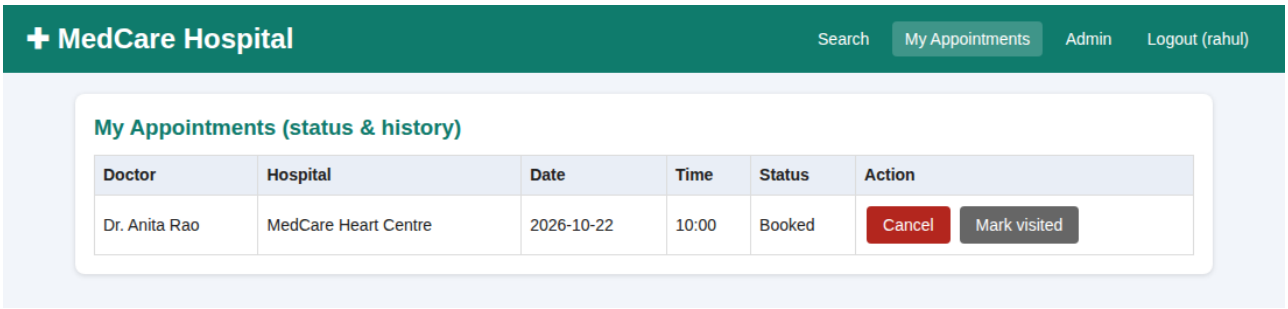
Search module - list of doctors with ratings



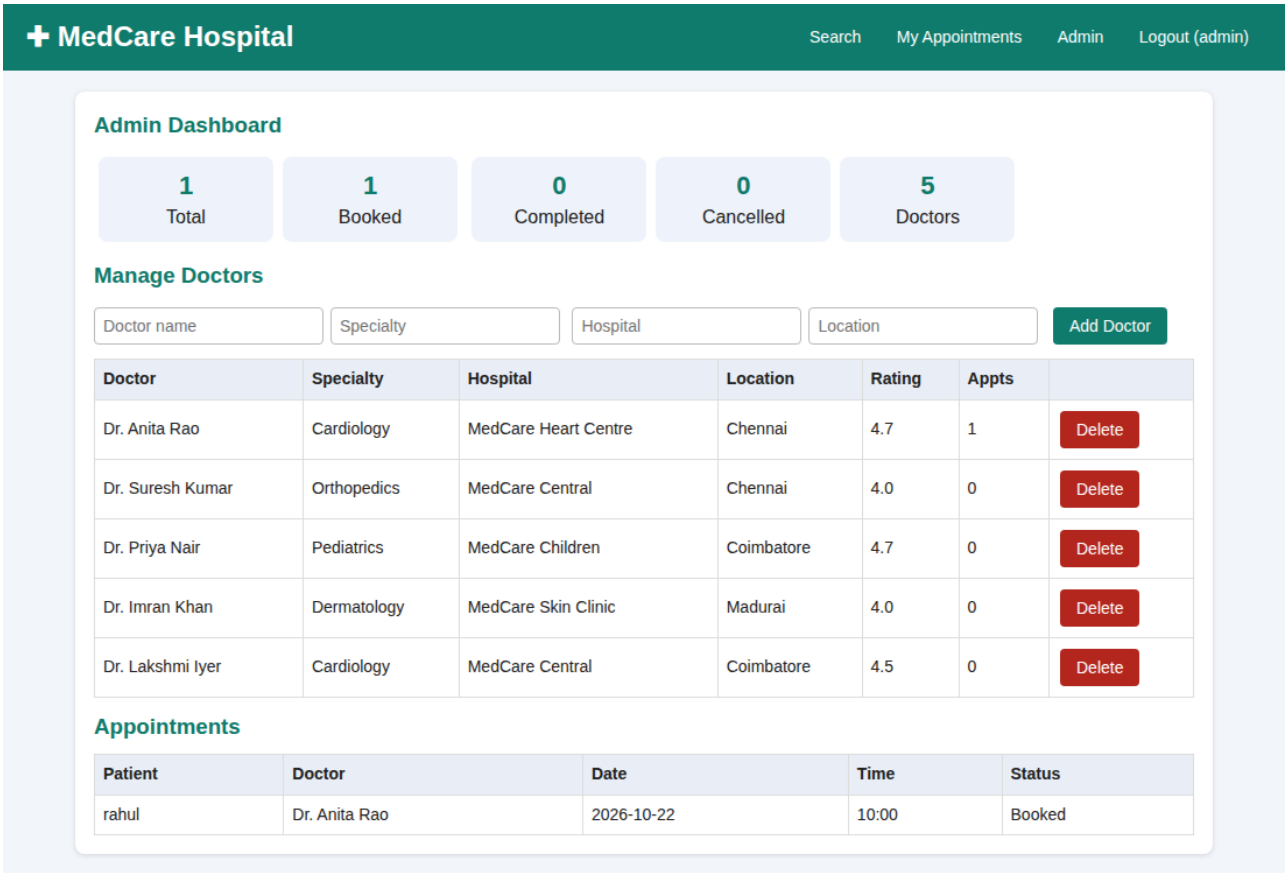
Search filtered by specialty (Cardiology)



Appointment booking - slot selection and confirmation message



My Appointments - appointment management (Cancel / Mark visited)



Admin Dashboard - statistics, manage doctors and appointments

RESULT

Thus the hospital online appointment booking system was designed and implemented using HTML, CSS and JavaScript. Patients could search doctors, book and manage appointments and the administrator could manage doctors and view statistics. The output was verified successfully.